

YAHIA HAMMAD

Software Engineer · ML & Full-Stack · Third-year CS @ MIU

+20 112 091 6055 · Yahiamhammad@gmail.com · yahiahammad.vercel.app · Cairo, Egypt

EXPERIENCE

AI & Data Science Intern · PaySky

Jun – Aug 2025

- Built a complete PySpark ETL pipeline ingesting 24M+ transaction, user, and card records — reducing data prep from raw exports to analysis-ready datasets in a single automated run.
- Applied and benchmarked five clustering algorithms (K-Means, DBSCAN, GMM, Agglomerative, Rule-Based); selected Agglomerative Clustering based on silhouette scores and cluster interpretability.
- Segmented the user base into actionable marketing personas (Loyal, At-Risk, Churned) and layered geographic segmentation via an interactive Plotly Mapbox heat map.
- Deployed the full solution with a FastAPI backend, React.js frontend, and Docker containers, enabling the marketing team to explore segments in real time.

Teaching Co-Assistant · Misr International University

Sep 2024 – Jan 2025

- Led C++ tutorial sessions and provided one-on-one support for a cohort of first-year students, directly improving lab pass rates.

PROJECTS

Wireless Gesture-Controlled Robotic Arm *ESP32 · Arduino · MPU-6050 · PCA9685 · C++* Spring 2026

- Engineered a tendon-driven robotic hand and wearable sensor glove achieving 1:1 real-time motion replication over low-latency Wi-Fi UDP between dual ESP32s.
- Designed the actuation system with a PCA9685 I2C controller driving MG996R/MG90S servos via zero-stretch braided lines through PTFE tubing for precise finger kinematics.
- Built a wearable PCB integrating an MPU-6050 IMU and custom-soldered flex sensors; implemented dual power architecture separating motor and LiPo supplies for stable telemetry.

CareerPath AI — Career Development Platform *Next.js · TypeScript · Supabase · Groq (Llama 3.3) · RAG* Fall 2025

- Architected a RAG pipeline using Groq SDK and Supabase Vector Store with Xenova Transformers for semantic search, delivering personalized course recommendations and automated skill extraction.
- Built an interactive career counselor chatbot that visualizes career trajectories and pivot paths using Mermaid.js diagrams generated on-the-fly by the LLM.
- Secured user data with PostgreSQL Row Level Security policies and a custom Fuse.js assessment engine mapping user proficiencies against industry benchmarks.

Unibite — University Food Ordering System *Node.js · Express.js · MongoDB · EJS · JWT* Fall 2025

- Delivered a full-stack campus food ordering platform supporting student/staff authentication, role-based access, real-time order management, and mobile-responsive UI.
- Hardened the backend with JWT security, CORS policies, rate limiting, stale-pricing guards, VAT tracking, and a cron job for auto-completing abandoned orders.

AI Fitness Tracker *Python · OpenCV · MediaPipe*

Fall 2025

- Built a real-time posture correction and rep-counting system using trigonometric joint-angle computation over live video; applied frame-smoothing to reduce landmark noise.

TECHNICAL SKILLS

Languages: Python, JavaScript / TypeScript, Java, C++, C
Frontend: React, Next.js, Tailwind CSS, HTML/CSS, ShadCN, Radix UI, Recharts
Backend & DB: Node.js, Express.js, FastAPI, MongoDB, PostgreSQL, Supabase, MySQL
AI / ML: PySpark, Scikit-learn, TensorFlow, Pandas, NumPy, RAG, Groq SDK, OpenCV, MediaPipe
Tools: Docker, Git, Plotly, Matplotlib, Seaborn, Cisco Packet Tracer

EDUCATION

BSc Computer Science · Misr International University 2023 – 2027 · GPA 3.32 / 4.0
Supervised Machine Learning: Regression & Classification 2024
DeepLearning.AI & Stanford University — Coursera · Linear/logistic regression, gradient descent, regularisation, NumPy, Scikit-learn
Digital Egypt Pioneers — Data Science Scholarship 2024 – 2025
Egyptian Ministry of Communications & IT · Data analysis, ML, big data — Pandas, NumPy, TensorFlow, Scikit-learn
Languages: English (C1 — Professional) · Arabic (Native)